BUG CATCHING GAME

AIM

The aim of the Bug Catcher game is to develop a simple and interactive game using Python. The player must catch bugs that appear on the screen by clicking on them with the mouse. Each successful catch increases the score. The game uses animation to make the bugs move around the screen. The main objective is to catch as many bugs as possible before the game ends.

PROCEDURE

1. Start the program and import the required Python libraries for creating the game window, graphics, random bug positions, and mouse controls.

2. Create the main game window using Tkinter and set the title as "BUG CATCHER GAME".

3. Design the game board with a background, a bug, a score display, a timer, and a Start / Restart button.

4. Set the initial score to zero and the game timer to 30 seconds.

5. Generate the bug at a random position on the game board using the random module.

6. Implement mouse-click control so the player can catch the bug by clicking on it with the left mouse button.

7. Animate the bug by changing its position at regular intervals using the after() function.

8. Detect successful clicks on the bug, remove the caught bug, and increase the player's score.

9. Update the timer every second and display the final score when the time reaches zero.

10. Run and test the game to verify bug movement, mouse-click detection, score updates, timer operation, and the game-over message.

PROGRAM CODE

```
import tkinter as tk
 import random

 WIDTH = 700
 HEIGHT = 450
 GAME_SECONDS = 30

 root = tk.Tk()
 root.title("Bug Catcher")
 root.resizable(False, False)

 score = 0
 time_left = GAME_SECONDS
 running = False
 bug_id = None
 timer_job = None
 move_job = None

 canvas = tk.Canvas(root, width=WIDTH, height=HEIGHT, bg="#dff7d8", highlightthickness=0)
 canvas.pack()


score_text = canvas.create_text(85, 25, text="Score: 0", font=("Arial", 16, "bold"), fill="#245b35")
 time_text = canvas.create_text(610, 25, text="Time: 30", font=("Arial", 16, "bold"), fill="#245b35")
 message_text = canvas.create_text(WIDTH // 2, HEIGHT // 2, text="Click Start to catch bugs!",
font=("Arial", 20, "bold"), fill="#245b35")

 def remove_bug():
 global bug_id
 if bug_id is not None:
 canvas.delete(bug_id)
 bug_id = None
 canvas.delete("bug_face")
```

PROGRAM CODE(continued)

```
def move_bug():
 global bug_id, move_job
 if not running:
 return
 remove_bug()
 x = random.randint(35, WIDTH - 35)
 y = random.randint(75, HEIGHT - 35)
 bug_id = canvas.create_oval(x - 20, y - 20, x + 20, y + 20,
 fill="#ef476f", outline="#8b1e3f", width=2,
 tags="bug")
 canvas.create_text(x, y, text="🐞", font=("Arial", 19), tags=("bug_face",))
 # Bind the visible bug and its face to the same click action.
 move_job = root.after(1100, move_bug)

 def catch_bug(event=None):
 global score
 if running and bug_id is not None:
 score += 1
 canvas.itemconfigure(score_text, text=f"Score: {score}")
 remove_bug()
 if move_job:
 root.after_cancel(move_job)
 move_bug()

def countdown():
 global time_left, running, timer_job
 if not running:
 return
 time_left -= 1
 canvas.itemconfigure(time_text, text=f"Time: {time_left}")
 if time_left <= 0:
 running = False
 remove_bug()
 canvas.itemconfigure(message_text, text=f"Time's up! Final score: {score}")
 else:
 timer_job = root.after(1000, countdown)

def start_game():
 global score, time_left, running, timer_job, move_job
 if timer_job:
 root.after_cancel(timer_job)
 if move_job:
 root.after_cancel(move_job)
 score = 0
```

PROGRAM CODE(continued)

```
time_left = GAME_SECONDS
 running = True
 canvas.itemconfigure(score_text, text="Score: 0")
 canvas.itemconfigure(time_text, text=f"Time: {GAME_SECONDS}")
 canvas.itemconfigure(message_text, text="")
 remove_bug()
 move_bug()
 countdown()

 def on_click(event):
 # Find whether the click is on the current bug.
 if bug_id is not None:
 coords = canvas.coords(bug_id)
 if coords and coords[0] <= event.x <= coords[2] and coords[1] <= event.y <= coords[3]:
 catch_bug()

 canvas.bind("<Button-1>", on_click)

 controls = tk.Frame(root, bg="#f4fff0")
 controls.pack(fill="x")
 tk.Button(controls, text="Start / Restart", command=start_game,
 font=("Arial", 12, "bold"), bg="#80c783", fg="#173b24",
 padx=12, pady=6).pack(pady=8)

 root.mainloop()
```


Bug Catcher
BUG CATCHER
Catch the bugs, get points!
Start
Instructions


Bug Catcher
Score: 3
Time: 22
Restart


Bug Catcher
Score: 4
Time: 18
Bug Caught!
+1 Point
Restart


Bug Catcher
Time's Up!
Final Score:
12
Play Again
Main Menu

RESULT

Thus, the Bug Catcher Game was implemented using Python Tkinter. The game provides a graphical playing area, randomly positioned bugs, mouse-click controls, score tracking, and a 30-second timer. The player can catch bugs by clicking on them and increase the score. The game demonstrates GUI development, animation, random positioning, mouse events, and basic game logic. At the end of the game, the final score is displayed.